

The Limit Superior and Limit Inferior

Your Reliable Ally

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This article aims to demistify the concept of $\limsup$ and $\liminf$ which students have to learn in an introductory analysis course right after learning about the rigorous definition of convergence in $\mathbb{R}$. We will build this concept from ground up so the idea feels natural to the students.

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1. Reviewing Convergence in Reals

We'll start by recalling the definition of convergence of a real sequence a_n in $\mathbb{R}$.

Definition 1.1 (Convergence in $\mathbb{R}$). We say that a sequence a_n converges in $\mathbb{R}$ if there exists some $a \in \mathbb{R}$ such that for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon \quad \text{for any } n > N.$$

The idea for this definition is quite simple, for large enough N , the distance between the terms of the sequence a_n and a can get arbitrarily small. Essentially, we are *analyzing* the **long-term ‘settling’ behaviour of the sequence**.

This definition has some problems. First of all, if such an a exists, then is it unique? Does it even exist? The answer to the first problem is given by the following theorem.

Theorem 1.2 (Uniqueness)

If a real sequence a_n converges to both a and a' , then $a = a'$. In other words, the limit of the sequence is unique.

Due to the above theorem, we can now introduce a notation. If a_n converges to a , then we shall write it as $\lim_{n \rightarrow \infty} a_n = a$ or $a_n \rightarrow a$ as $n \rightarrow \infty$.

If you consider a sequence $a_n = (-1)^n$, then it can be proven that a_n doesn't converge to any real number. So, given any sequence, it is not necessary that the sequence must converge.

The majority of real analysis, deals with convergent sequences. The sequences which don't converge to anything, treating them becomes harder. Here's a theorem which has extensive use throughout analysis.

Theorem 1.3 (Limit Inequality)

Let a_n and b_n be two real sequences such that $a_n \geq b_n$ and both the sequences are convergent in $\mathbb{R}$. Then

$$\lim_{n \rightarrow \infty} a_n \geq \lim_{n \rightarrow \infty} b_n$$

The theorem has a *strong hypothesis*; it assumes the convergence of both the sequences. If the sequences are not convergent, then what shall we do? We have now finally hit the grave the problem, if the sequence ceases to converge, then is there any useful notion to find where the sequence is *settling* in the long run?

1.1. A Grave Problem

Example 1.4 (A Really (Un?)Settling Sequence)

Consider the sequence $\{a_n\}_{n \geq 1}$ given by

$$a_n = (-1)^n \left(2 + \frac{1}{n}\right).$$

Is a_n convergent? Does it *seem to converge* anywhere?

For the beginners, it is a good exercise to sit down and write a formal proof that the sequence doesn't converge. Although, if you take large n , then one realizes that a_n seems to sometimes approach 2 but sometimes approaches -2 . Here's a small table, for values of n around 1000.

n	a_n
997	$-2.001003\dots$
998	$2.001002\dots$
999	$-2.001001\dots$
1000	$2.001000\dots$
1001	$-2.000999\dots$
1002	$2.000998\dots$
1003	$-2.000997\dots$

Formally, we can write it as

$$\lim_{n \rightarrow \infty} a_{2n} = 2 \quad \& \quad \lim_{n \rightarrow \infty} a_{2n-1} = -2.$$

This means that even though the sequence doesn't converge (due to [Theorem A.3](#)), there is still some settling behaviour in the sequence. To analyze the long term 'settling' behaviour, we definitely want to capture this behaviour.

1.2. Important Terminologies

With the motivation set in space, we'll now introduce a new definition.

Definition 1.5 (Accumulation Points). For a real valued sequence $\{a_n\}_{n \geq 1}$, we say that u is an *accumulation point* if for any $\varepsilon > 0$ there exists infinitely many $m \in \mathbb{N}$ such that

$$a_m \in (u - \varepsilon, u + \varepsilon).$$

In essence, this definition says that there are infinitely members of sequence a_n which are lying arbitrarily arbitrarily close to u . A similar definition which textbooks use is the following.

Definition 1.6 (Subsequential Limits). We say u is a subsequential limit of sequence $\{a_n\}_{n \geq 1}$ if there exists some subsequence $\{a_{n_k}\}_{k \geq 1}$ such that $a_{n_k} \rightarrow u$ as $n \rightarrow \infty$.

It's not a surprise that any accumulation point must be a subsequential limit of the sequence. Think like this, any subsequential limit **MUST** be an accumulation point by very definition of convergence. If you have accumulation point in hand, then first pick an element within 2^{-1} neighborhood of u . Then pick a new element from 2^{-2} neighborhood. Rinse and repeat!

This way one can construct a subsequence converging to u . Details are obviously missing. Thus, we shall formally state it and prove it as a theorem.

Theorem 1.7

$v \in \mathbb{R}$ is an *accumulation point* if and only if v is a *subsequential limit*.

Proof. To be Added. . . □

With help of above theorem, we can now use the words *accumulation points* and *subsequential limits* interchangeably. The following theorem becomes quite trivial.

Theorem 1.8

A sequence $\{a_n\}$ converges to $a \in \mathbb{R}$ if and only if $\{a_n\}_{n \geq 1}$ has only a unique *accumulation point*, a to be particular.

Exercise 1.9 (Optional). Write down formal proof to the above theorem if you are a beginner or look it up if you are feeling stuck.

The idea of $\limsup$ is exactly the same as finding the “largest” accumulation point. Its a good time to revisit [Example 1.4](#).

We could clearly see that a_n was kind of settling around the points 2 and -2 . Indeed, due to [Theorem 1.7](#) both 2 and -2 are *accumulation points*. It requires some work to show that there are no other accumulation points other than 2, -2 .

Exercise 1.10. Show that the sequence

$$a_n = (-1)^n \left(2 + \frac{1}{n} \right)$$

has only two *accumulation points*, which are 2 and -2 .

The largest *accumulation point* of a_n is 2, thus we would write $\limsup a_n = 2$. There are some technical details which one needs to address, like its existence and whether a true maximum¹ exists or not. The next section is dedicated to properly establishing this notion and filling any holes of this concept which we have in mind.

2. Introducing Limit Superior

We’ll walk through a few equivalent definitions and give the proper proofs wherever necessary. We will start with the subsequential limits definition and then move onto the more *harder* definition which allows for easier computation of the limit superior.

Note: All the sequences are **bounded** unless stated otherwise.

¹if you consider a set of all accumulation points, it could be the case a *supremum* exists, but a maximum doesn’t.

2.1. Largest Accumulation Point/Subsequential Limit

We have now seen that the limit of the sequence is indeed an *accumulation point*. The idea of limit supremum is finding the largest *accumulation point* of your a_n . There is no guarantee whether a true ‘maximum’ exists or not. Therefore, for sequence a_n define

$$V := \{x \in \mathbb{R} : x \text{ is an accumulation point of } a_n\}.$$

We want to find $\sup(V)$, for which we shall use [The Completeness Axiom](#). We need to ensure that V is bounded above and non-empty. The following lemma will do it.

Lemma 2.1

V is bounded above and non-empty.

Proof. Let $u \in \mathbb{R}$ be an *accumulation point* or a *subsequential limit*. Thus, there exists some subsequence $\{a_{n_k}\}_{k \geq 1}$ such that $a_{n_k} \rightarrow u$ as $k \rightarrow \infty$. Since a_n is bounded, there exists some $M > 0$ such that $-M \leq a_n \leq M$ for all $n \geq 1$. Therefore,

$$-M \leq a_{n_k} \leq M \quad \text{for all } k \geq 1.$$

Taking $m \rightarrow \infty$ on both the sides with aid of [Theorem 1.3](#), we shall $|u| < M$ for any accumulation point u of a_n , proving that V is **bounded**.

Since a_n is bounded, we know that via [Bolzano Weirstrass](#), there exists a subsequence $\{a_{m_k}\}_{k \geq 1}$ such that $a_{m_k} \rightarrow t$ for some $t \in \mathbb{R}$. Since t is a *subsequential limit*, it must be an *accumulation point*, which shows $t \in V \implies V \neq \emptyset$. $\square$

With this preparatory lemma, we are now free to define the following.

Definition 2.2. For a bounded real sequence $\{a_n\}_{n \geq 1}$, define

$$\limsup_{n \rightarrow \infty} a_n = \sup(V)$$

where V is set of all accumulation points of $\{a_n\}_{n \geq 1}$.

Remark (Notation). There are several notations for this quantity. People write it as $\overline{\lim} a_n$ or just $\limsup a_n$. We shall justify it later why we use $n \rightarrow \infty$ below $\limsup$ in a latter section.

We have finally setup the definition of the limit superior. A question still remains... Is the supremum attainable? The answer turns out to be a **yes**. The following lemma is from Rudin but is very crucial to get a better understanding about the ‘distribution’ of *accumulation points* on the real line.

If a student wants to freshen up on the concept of what a ‘closed’ set is, then they can refer **here** in the appendix.

Lemma 2.3

The *subsequential limits* of a real sequence $\{p_n\}_{n \geq 1}$ forms a closed subset of $\mathbb{R}$.

Proof. To be added...

□

Exercise 2.4. Prove that the supremum of a bounded closed set exists inside the set. Therefore, show that $\sup(V) \in V$.

Exercise 2.5 (General Metric Spaces). Write a general result, replacing $\mathbb{R}$ by a general metric space (X, d) where X is the set of elements and d is the distance function. Write a formal proof as well.

This definition is a little difficult to handle. Finding accumulation points in itself takes quite some computation and then we are here talking about the ‘largest one’. In particular, we want some properties of this number which are actually useful in practice.

Regardless, this is simplest definition to introduce to a student initially. Instead of directly stating another equivalent definition, we’ll take sometime analyzing what kind of behaviour should our ‘largest’ *accumulation point* exhibit.

2.2. The Epsilon Criteria

In this subsection, we will carry-forward and build up more on the previous definition of limit superior. We wish to analyze what kind of behaviour does the largest *accumulation point* exhibits.

What happens when you nudge the $\limsup$ slightly? Let $s^* = \limsup a_n$, what happens if you consider which is slightly greater than s^* ? For now, $y = s^* + \epsilon$ for some $\epsilon > 0$. Since it is greater than $\sup(V)$, y is definitely **NOT** an *accumulation point*.

Recall via the proof of V being non-empty and bounded, if there are infinitely many terms greater than y , then via [Bolzano Weirstrass](#), we will get an *accumulation point* strictly greater than s^* , which is absurd since s^* is the maximum such real. Thus, we may conclude that there are only finitely many $n \in \mathbb{N}$ such that $y = s^* + \epsilon > a_n$.

What if you nudge the s^* backwards? This time, let $z = s^* - \epsilon$. By the very definition of $\sup(V)$, there exists some $v \in V$ such that $z < v \leq s^*$. From here, it’s easy to conclude that there are infinitely many n such that $z = s^* - \epsilon < a_n$. All our discussion till now, can be summarized in the following theorem.

Theorem 2.6

For any bounded real sequence $\{a_n\}_{n \geq 1}$, let $s^* = \limsup_{n \rightarrow \infty} a_n$, then

- for any $\epsilon > 0$, there are finitely many n satisfying $s^* + \epsilon < a_n$;
- for any $\epsilon > 0$, there are infinitely many n satisfying $s^* - \epsilon < a_n$.

Proof. To be added...

□

The above theorem is **extremely useful** when dealing with $\limsup$ or $\liminf$ (with the appropriate changes). The above two properties are quite unique, in the sense only the $\limsup$ satisfies both the above properties together. Let's state this "uniqueness" property clearly.

Lemma 2.7

For a bounded sequence $\{a_n\}_{n \geq 1}$, if $s^* \in \mathbb{R}$ such that

- For any $\varepsilon > 0$, there are finitely many n satisfying $s^* + \varepsilon < a_n$;
- For any $\varepsilon > 0$, there are infinitely many n satisfying $s^* - \varepsilon < a_n$.

Then there is only one $s^* \in \mathbb{R}$ which satisfies both of these properties.

Proof. To be added...

□

Magic! We already know that $\limsup a_n$ satisfies both of the properties and due to the above lemma, we now know $\limsup a_n$ is the only number which can satisfy both the properties together. Therefore, the concept of $\limsup$ could be purely defined in terms of this ε -criteria. We won't do that here, but the theorem stated above will be a very important result which will be consistently used in proofs regarding $\limsup$ and $\liminf$ later on.

Exercise 2.8 (Optional). Write down the definition of $\limsup_{n \rightarrow \infty} a_n$ purely in terms of the ε -criteria. Note that, the definition is completely useless unless you actually find a suitable candidate for s^* .

2.3. Tailed Supremums

This definition of $\limsup$ is arguably the most important definition and what is mostly presented in textbooks. We'll try to give some motivation, as to why would one come up with such a definition in the first place.

2.4. The Limit Inferior

We have not given any attention to limit inferior in our discussion. Limit Inferior has completely the same idea as limit superior, but this time we want to look at the 'smallest' *accumulation point*.

All the results that were true for $\limsup$ are true with $\liminf$, with only slight modifications. Sometimes we might have to change the sign of the inequality, replace the word 'greatest' with 'smallest'.

The attentive readers should have no problem in deciphering the concept of limit inferior. Thus, we leave it as an exercise.

Exercise 2.9 (Limit Inferior). Try to reconstruct the entire article upto Section 2, but instead of talking about the $\limsup$, the reconstructed article should focus on $\liminf$ as the main subject.

2.5. Useful Results

Theorem 2.10

For any real valued bounded sequence $\{a_n\}_{n \geq 1}$,

$$\limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n = a \iff \lim_{n \rightarrow \infty} a_n = a.$$

Theorem 2.11 (Monotonicity)

Let $\{a_n\}_{n \geq 1}$ and $\{b_n\}_{n \geq 1}$ satisfy $a_n \geq b_n$ for all $n \geq 1$. Then

$$\begin{aligned} \limsup_{n \rightarrow \infty} a_n &\geq \limsup_{n \rightarrow \infty} b_n \\ \liminf_{n \rightarrow \infty} a_n &\geq \liminf_{n \rightarrow \infty} b_n \end{aligned}$$

2.6. Heuristics: When to use it?

There are quite a few applications.

1. One can solve problems related to finding the limit superior and limit inferior. This sounds artificial but it is definitely an application.
2. $\limsup$ and $\liminf$ always exist! So, whenever we are solving a problem, we can't apply theorems like [Theorem 1.3](#). Although, any theorem regarding $\limsup$ and $\liminf$ can be applied almost everywhere!
3. This is a very useful tool for showing that the limit exists, courtesy to [Theorem 2.10](#). This tool which we have developed can be replaced by first principles (some $\varepsilon - \delta$ arguments) rather than analyzing the *accumulation points*. Although, thinking in terms of *accumulation points* hugely simplifies our proofs and is a good direction to think in.
4. The next section will present some theorems where this new notion of $\limsup$ makes stating these results extremely easy. The root and ratio tests for convergence of series is a very good example which we shall cover later.

To put it in a few words, *if you don't know whether the limit exists or not, then take the help of your best ally, the limit superior.*

3. Brief Note on Extended Reals and Unbounded Sequences

Throughout the article, we have only dealt with bounded sequences. What happens if the sequence is unbounded? The ‘largest’ *accumulation point* of a sequence breaks down... If you try to imitate the proof [Lemma 2.1](#), showing V bounded is not possible since a subsequence might converge to $\pm\infty$.

Therefore, if the sequence is unbounded we would like to work with extended reals, which is

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{+\infty, -\infty\}$$

When working with extended reals, we know that $\sup(V)$ always exists (just need to ensure $V \neq \emptyset$) in $\overline{\mathbb{R}}$. There are quite a few conventions and definitions one needs to know when working with extended reals, but the following will suffice.

1. $0 \times \pm\infty$ and $\pm\infty \times 0$ are undefined.
2. $\infty - \infty$ is undefined.
3. $\infty + a = a + \infty = \infty$ for all $a \in \mathbb{R}$.
4. $a + (-\infty) = (-\infty) + a = -\infty$ for $a \in \mathbb{R}$.

The discussion for unbounded sequences is rather trivial, we’ve dealt with the non-trivial part already. Therefore, for brevity here is a sketch how to move from one definition to the other.

- Hello.

4. Few Applications

4.1. Artificial Problems

4.2. The Ratio and Root Tests

Theorem 4.1 (Root Test)

Consider a real-valued sequence $\{a_n\}_{n \geq 1}$. Define

$$\alpha := \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

Then

1. if $\alpha < 1$, $\sum a_n$ converges;
2. if $\alpha > 1$, $\sum a_n$ diverges;
3. if $\alpha = 1$, the test is inconclusive.

Moreover, whenever $\alpha < 1$, the convergence is *absolute*.

Proof. To be added...

□

Theorem 4.2 (Ratio Test)

For a real-valued sequence $\{a_n\}_{n \geq 1}$, with $a_n \neq 0$ for $n > N$ for some $N \in \mathbb{N}$, define

$$\beta := \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

Then

1. if $\beta < 1$, $\sum a_n$ converges;
2. if $\beta > 1$, $\sum a_n$ diverges;
3. if $\beta = 1$, the test is inconclusive.

Moreover, whenever $\beta < 1$, the convergence is *absolute*.

Proof. To be added...

□

4.3. Radius of Convergence of Power Series**4.4. An Example from Probability Theory**

This one is too niche, regardless a good example. If one is not aware of the notion of weak convergence/convergence in distribution in probability theory, then feel free to skip this one for now.

Problem (No Atoms in Weak Convergence)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{X_n\}_{n \geq 1}$ be a sequence of real valued-random variables and X be another random variable such that $X_n \xrightarrow{d} X$. If a is a continuity point of F_X , then prove that

$$\lim_{m \rightarrow \infty} \mathbb{P}(X_m = a) = \mathbb{P}(X = a) = 0.$$

Solution. Let F_n denote the CDF of X_n . Since F_n is monotonically increasing, there are only countable discontinuities of F_n . Therefore, we may choose a strictly decreasing sequence $\{\delta_m\}_{m \geq 1}$ approaching 0 such that $a - \delta_m$ and $a + \delta_m$ are both continuity points of F_X for all $m \geq 1$. Observe,

$$\begin{aligned} \mathbb{P}(X_n = a) &\leq \mathbb{P}(X_n \in (a - \delta_m, a + \delta_m]) \\ &= F_n(a + \delta_m) - F_n(a - \delta_m) \end{aligned}$$

Since $a + \delta_m$ and $a - \delta_m$ were chosen to be continuity points of F_X , the limit as $n \rightarrow \infty$ of $F_n(a + \delta_m)$ and $F_n(a - \delta_m)$ exists and is precisely $F_X(a + \delta_m)$ and $F_X(a - \delta_m)$. Taking $\limsup_{n \rightarrow \infty}$ on both the sides with help of [Theorem 2.10](#), we get

$$\limsup_{n \rightarrow \infty} \mathbb{P}(X_n = a) \leq F_X(a + \delta_m) - F_X(a - \delta_m)$$

for all $m \geq 1$. Once again, since $a + \delta_m$ and $a - \delta_m$ are continuity points of F_X , if we take $m \rightarrow \infty$ on both the sides, $F_X(a + \delta_m) \rightarrow F_X(a)$ and $F_X(a - \delta_m) \rightarrow F_X(a)$ since a is a point of continuity. This yields

$$\limsup_{n \rightarrow \infty} \mathbb{P}(X_n = a) \leq 0 \implies \limsup_{n \rightarrow \infty} \mathbb{P}(X_n = a) = 0$$

Since $\mathbb{P}(\bullet) \geq 0$, taking $\liminf$ both sides gives

$$0 \leq \liminf_{n \rightarrow \infty} \mathbb{P}(X_n = a) \leq \limsup_{n \rightarrow \infty} \mathbb{P}(X_n = a) = 0,$$

forcing both $\limsup$ and $\liminf$ to be equal to 0, which proves

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = a) = 0$$

completing the proof. ■

5. Limit Superior and Limit Inferior of Sets

Being a student at ISI Kolkata, I feel the most natural application to this concept is in Probability Theory. We shall discuss a few examples later, for now we shall only consider examples which are more related to the real line.

5.1. Connection with Limit Superior/Inferior of Sequences

5.2. Applications in Probability Theory

6. Practice Problems

Problem 6.1 (Alternate Definition). Let $\{x_n\}_{n \geq 1}$ be a bounded real sequence. We now define

1. The limit superior of x_n is the infimum of set V of $v \in \mathbb{R}$ such that $v < x_n$ for at most a finite number of $n \in \mathbb{N}$.
2. The limit inferior of x_n is the supremum of the set W of w such that $x_m < w$ for at most a finite number of $m \in \mathbb{N}$.

Show that the above definition is the same as tail-end definition of $\limsup$ which we have discussed earlier.

Problem 6.2 (First Borel-Cantelli Lemma). Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a sequence events $\{A_n\}_{n \geq 1}$ where $A_m \in \mathcal{F}$ for all $m \in \mathbb{N}$, then prove that

$$\sum_{i=1}^{\infty} \mathbb{P}(A_i) < \infty \implies \mathbb{P}\left(\limsup_{n \rightarrow \infty} A_n\right) = 0.$$

A. Appendix

A.1. Standard Facts from Analysis-I

Definition A.1 (Limit Points). $u \in \mathbb{R}$ is said to be a *limit point* of $A \subseteq \mathbb{R}$ if for any $\varepsilon > 0$, there exists some $u^* \in A$ such that

$$u^* \in (u - \varepsilon, u + \varepsilon) \setminus \{u\}.$$

Definition A.2 (Closed Sets). A subset $A \subseteq \mathbb{R}$ is said to be a *closed set* if it contains all its limit points.

Axiom (The Completeness Axiom)

For any non-empty $A \subset \mathbb{R}$ bounded above, the supremum of A or $\sup(A)$ exists in $\mathbb{R}$.

Theorem A.3 (Subsequences of Convergence Sequences)

If $a_n \rightarrow a$ as $n \rightarrow \infty$, then any subsequence of a_n must converge to a as well.

Theorem A.4 (Bolzano Weirstrass)

For any bounded sequence $\{b_n\}_{n \geq 1}$, there exists a convergent subsequence $\{b_{n_k}\}_{k \geq 1}$.